

Code-Layout, Readability and Reusability

Date	12 August 2026
Name	Chahat Chouhan
Team ID	Individual Project
Project Name	AI-Brand-Voice-Generator
Maximum Marks	5 Marks

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Adhered strictly to PEP 8 standards (4 spaces) for Python backends and Prettier formatting for frontend files.
2	Proper File Structure	Files and folders are logically organized	Yes	Standard MVC/modular pattern used (<code>/routes</code> , <code>/services</code> , <code>/components</code> , <code>/models</code> , <code>/utils</code>).
3	Meaningful	Variables	Yes	Descriptive

	Variable Names	reflect their purpose clearly		naming conventions followed (e.g., <code>extracted_brand_tone</code> , <code>user_auth_token</code>).
4	Function / Method Names	Functions are descriptively named	Yes	Action-oriented verbs used (e.g., <code>extract_tone_profile()</code> , <code>generate_marketing_copy()</code>).
5	Code Comments	Inline and block comments explain logic	Yes	Docstrings added for API endpoints and complex prompt-building algorithms.
6	Modular Design	Code is split into reusable functions/modules	Yes	Business logic decoupled from API routing into dedicated service helper functions.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Refactored shared validation schemas and utility wrappers to eliminate boilerplate repetition.
8	Error Handling	Exceptions and errors are handled	Yes	Global exception handlers

		gracefully		wrap LLM call failures and validation errors with clean HTTP status responses.
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Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Gemini API Service Wrapper	Python (FastAPI)	Tone Extraction Endpoint & Copy Generation Engine	High
2	JWT Authentication Middleware	Python / FastAPI Security	Across all protected API routes (/api/generate, /api/profile)	High
3	Brand Profile Form Component	React.js / JavaScript	User Dashboard & Admin Config Page	Medium

4	Copy Output Card Component	React.js / Tailwind CSS	Generation Results Screen & Search History Log	Medium
5	Input Text Validator / Sanitizer	Python (Pydantic)	Pre-processing incoming request payloads before LLM ingestion	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5 / 5	Clean separation between frontend components, backend routes, and database models.
Readability	4.5 / 5	Highly readable syntax using standard naming conventions and consistent formatting toolings.
Reusability	4.5 / 5	Core LLM integrations and UI elements are modularized effectively for scalable expansion.
Documentation / Comments	4.5 / 5	Comprehensive docstrings for backend functions and clear README setup instructions.
Overall Score	4.6 / 5	Excellent Quality — The codebase is maintainable, scalable, and adheres to software engineering best practices.